Heat Equation

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Numerical solution of the steady-state heat equation in a metallic bar using the Finite Element Method (FEM).

It will be explored how heat propagates along a metallic bar of length L. The heat reservoir is at temperature T_R and is coupled to the bar at $x = 0$. This system is in contact with the surrounding environment, at ambient temperature T_A. The temperature of the bar is given by $T(x)$, with $x \in [0, L]$, and the equation for the temperature variation is

$$\frac{\partial u(x,t)}{\partial t} = \alpha \frac{\partial^2 u(x,t)}{\partial x^2} - \mu\, u(x,t), \tag{1}$$

where $u(x,t) = T(x,t) - T_A$ and μ is the loss rate to the atmosphere. Focusing on the stationary case (independent of t), one obtains

$$\frac{\partial^2 u(x)}{\partial x^2} - k\, u(x) = 0, \tag{2}$$

where $k = \mu/\alpha$. After applying the boundary conditions $T(0) = T_R = 2T_A$ and $T(L) = T_A$, and imposing $T_A = 1$, $L = 10$ and $k = 1$, the analytical solution is given by

$$u(x) = T_A\, \frac{e^{\sqrt{k}\,x} - e^{\sqrt{k}(2L-x)}}{1 - e^{20\sqrt{k}}} = \frac{e^{x} - e^{20-x}}{1 - e^{20}}. \tag{3}$$

I. INTEGRAL FORM

To solve this problem numerically, the Finite Element Method was used. This consists of discretizing the problem domain ($x \in [0, L]$) into $N+1$ nodes x_i, and approximating the solution using simpler functions associated with each node. The physics of the problem is then studied locally and, finally, all equations are assembled to obtain a global solution. Defining $u_i(x)$ as a triangular function for each node, and $\Delta x = L/N$ as the distance between nodes, one has

$$u_i(x) = \begin{cases} \dfrac{x - x_{i-1}}{\Delta x}, & x \in [x_{i-1}, x_i], \\ \dfrac{x_{i+1} - x}{\Delta x}, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

To further simplify the problem, the strong form given by (2) is transformed into the weak form of the equation. This is done by integrating over the entire domain the application of a test function v_j, required to project the residual error into a functional space, to equation (2), yielding:

$$\int_0^L \big[u''(x) - k\, u(x)\big]\, v_j(x)\, dx = 0. \tag{5}$$

The Galerkin Method is then applied, which defines the test functions as the same functions used in (5), i.e., $v_j = u_j$ and $j \in [0, N]$. This is a projection method that imposes orthogonality of the residual error to the approximation space through the weak form of the problem. Furthermore, if $u(x) \approx \sum_{i=0}^{N} a_i u_i$, then

$$\begin{aligned} &\sum_{i=0}^{N} a_i \int_0^L u_i''(x)\, u_j(x)\, dx = k \int_0^L u_i(x)\, u_j(x)\, dx \\ \Leftrightarrow\ &\sum_{i=0}^{N} a_i \big[u_i'(x)\, u_j(x)\big]_0^L - \sum_{i=0}^{N} a_i \int_0^L u_i'(x)\, u_j'(x)\, dx \\ &\qquad = k \int_0^L u_i(x)\, u_j(x)\, dx. \end{aligned} \tag{6}$$

Since $\big[u_i'(x)\, u_j(x)\big]_0^L = 0$, due to $u_j = 0$ at the boundaries, the integral (and weak) form of the problem is obtained:

$$\sum_{i=0}^{N} a_i \left[\int_0^L u_i'(x)\, u_j'(x)\, dx + k \int_0^L u_i(x)\, u_j(x)\, dx \right] = 0. \tag{7}$$

II. MATRIX FORM

Since each triangular function is equal to 1 at its respective node, one has $a_0 = 1$ and $a_N = 0$, as $u(0) = 1$ and $u(L) = 0$. Thus,

$$\begin{aligned} &\sum_{i=1}^{N-1} a_i \Big[\int_0^L u_i'(x)\, u_j'(x)\, dx + k \int_0^L u_i(x)\, u_j(x)\, dx \Big] \\ &\qquad = -a_0 \int_0^L u_0'(x)\, u_j'(x)\, dx. \end{aligned} \tag{8}$$

The problem can then be written in matrix form $A\vec{a} = \vec{b}$, obtaining a system of $N-1$ equations and unknowns.

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$\vec{a}$ corresponds to the vector of coefficients a_i, A_{ij} to the matrix entries $\left[\int_0^L u_i'(x)\, u_j'(x)\, dx + k \int_0^L u_i(x)\, u_j(x)\, dx\right]$, and $\vec{b}$ to the right-hand side of equation (8). From (5), one has

$$u_i'(x) = \begin{cases} \dfrac{1}{\Delta x}, & x \in [x_{i-1}, x_i], \\ \dfrac{-1}{\Delta x}, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise.} \end{cases}$$

Thus, the matrix A_{ij} is defined as follows:

$$A_{ij} = \begin{cases} \dfrac{2}{\Delta x} + \dfrac{2}{3}\Delta x, & \text{if } i = j, \\ -\dfrac{1}{\Delta x} + \dfrac{\Delta x}{6}, & \text{if } i = j \pm 1, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

From (10), it follows that $\vec{b}$ is zero except when $j = 1$:

$$b_j = \begin{cases} \dfrac{1}{\Delta x} - \dfrac{\Delta x}{6}, & \text{if } j = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

III. NUMERICAL SOLUTION

Using `Sparselib++`, the solution of the heat equation in its matrix form was obtained, yielding the vector $\vec{a}$. Since $u_i(x) = 1$ at the nodes, the analytical expression of the problem is compared with the value of a_i at each node $x = x_i$. For a tolerance of $\varepsilon = 10^{-8}$ and a maximum of 10^4 iterations, one has Observing Fig. 1, it is concluded

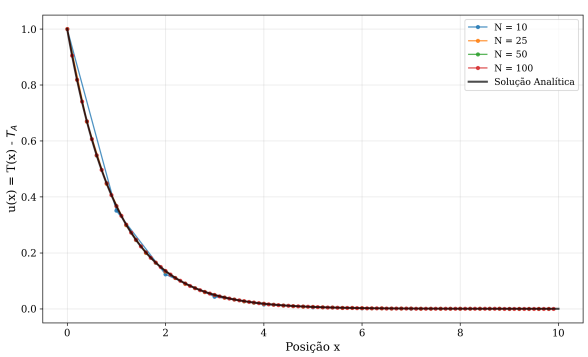

Figure 1: Comparison of the numerical solution for different discretizations with the theoretical one, for $N_{\text{iter}} = 10^4$, tolerance $\varepsilon = 10^{-8}$, $k = 1$.

that the larger the number of domain discretizations, the closer the numerical solution becomes to the analytical solution. Note that for $N = 10$ the graph does not show a good *fit* to the analytical solution, while for the remaining discretizations no relevant difference is observed relative to the theoretical solution. In this sense, the relative error plot for the same discretizations was also obtained, shown in Fig. 2. Examining Fig. 2, the previous statement is

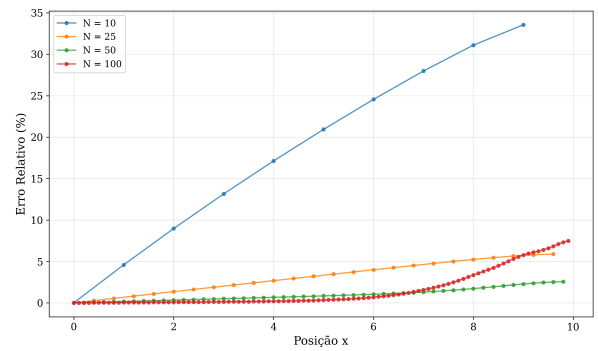

Figure 2: Relative error of different discretizations compared to the analytical solution, for $N_{\text{iter}} = 10^4$, tolerance $\varepsilon = 10^{-8}$, $k = 1$.

confirmed. It is observed that, for all values of N, there is a significant increase in error as $x \to L$. This is due to the fact that $u(L) = 0$, so when applying the relative error expression, values tend to diverge. Additionally, $u(L) = 0 < \varepsilon$, which worsens the situation. It is also noted that, although $N = 100$ shows smaller error over most of the domain, from $x = 7$ onwards the value becomes higher compared to $N = 50$ and $N = 25$. Thus, it is concluded that the most suitable discretization value is $N = 50$. Finally, the numerical solution was studied for different values of k. Analyzing Fig. 3, the variation of k reflects

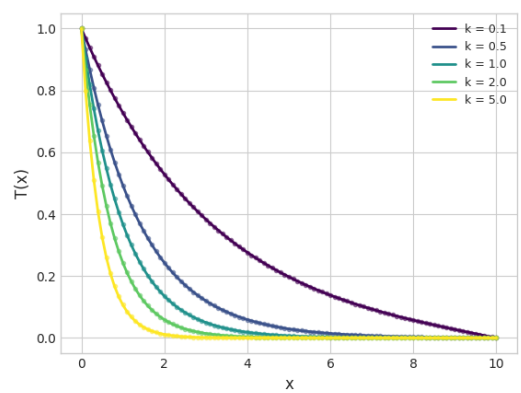

Figure 3: Solution computed using FEM for different values of k, with $N_{\text{iter}} = 10^4$, tolerance $\varepsilon = 10^{-8}$ and $N = 100$.

a proportional variation of μ, while keeping α constant (bars with the same thermal properties are considered). Higher values of k intensify heat losses and accelerate the decay of temperature along the bar, while lower values allow it to be maintained over longer distances.